

Xun Lei

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SUMMARY

Data Analyst with 3+ years expertise in **Machine Learning Technologies**, **Statistical Analysis**, and **Data Visualization**. Skilled in processing **large-scale datasets** and building practical **ML pipelines** including **data preprocessing**, **model training**, and **evaluation**. Passionate about transforming data into actionable insights and developing **predictive solutions** that enhance decision-making.

- **Programming:** Python (Pandas, NumPy, Scikit-learn, PySpark), SQL (3+ years), R (2+ years), Shell, Git
- **Data Visualization & BI Tools:** Svelte, JavaScript, Tableau, Matplotlib, Seaborn, CSS, HTML, ggplot
- **ML & Statistics:** Machine Learning (Regression, Clustering, Classification), Statistical Analysis, A/B Testing
- **Developer Tools & Platforms:** Spark, AWS S3 Buckets, EC2, MapReduce, Distributed Computing, Docker, Git, Shell

EDUCATION

Georgetown University | Washington, D.C. **Expected May, 2026**
Master of Science in Data Science and Analytics (GPA: 3.85/4.0)

- **Coursework:** Big Data & Cloud Computing, GIS for Spatial Data Science, Adaptive Measurement with AI

University of California, Santa Cruz (UCSC) | Santa Cruz CA **Sep. 2020 – Dec. 2023**
Bachelor of Science in Information Technology & Management (GPA: 3.7/4)

- **Coursework:** Database Systems, Financial Data Analysis, Cryptography, Algorithm Design

INDUSTRY EXPERIENCE

DATA SCIENCE INTERN **Apr. 2024 – June, 2024**
Qianji Technology Co.

- Developed and optimized AI models, utilizing **Stable Diffusion** for custom model training and **LoRA** fine tuning improving image generation accuracy by **20%**.
- Automated product image annotation and categorization using **Python**-based workflows, transitioning from manual **Excel** entries to an integrated pipeline while maintaining **95%** label consistency.
- Improved model generalization by **15%** through the implementation of stratified sampling and outlier detection, ensuring a balanced distribution of styles and perspectives in the training library.
- Expanded training datasets by reconstructing 3D product models, improving model adaptability and rendering consistency across diverse camera angles and complex lighting conditions.

TECHNICAL PROJECT

YouTube Video Popularity Analysis **Oct. 2024 – Dec. 2024**

- Analyzed YouTube video performance metrics including views, likes, comments, engagement rates, and duration using **YouTube API**, identifying key factors influencing video virality.
- Applied **K-Means** and **DBSCAN** clustering to categorize videos based on content topics and engagement metrics, uncovering audience behavior patterns and content preferences.
- Developed **Random Forest** and **Gradient Boosting models** to forecast video popularity trends, achieving an **18%** improvement in prediction accuracy over baseline linear models.
- Quantified the impact of like counts and duration through feature importance analysis, resulting in data-driven strategies to optimize video length that can drive a **25%** increase in audience retention.
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Large Scale Tech Community Engagement **Oct. 2025 – Dec. 2025**

- Distributed **Apache Spark (PySpark)** pipeline on **AWS EC2**, processing a **446 GB Reddit dataset** stored in **AWS S3** for large-scale community analytics.
- End-to-end processing latency reduced to seconds–minutes via **Kryo serialization**, **S3A performance tuning**, **Adaptive Query Execution (AQE)**, and **Parquet optimization**.
- Classified users into Creators and Participants, revealing that general interest subreddits generate ten times more comments per post and **39%** longer discussion threads than technical subreddits.